机器学习

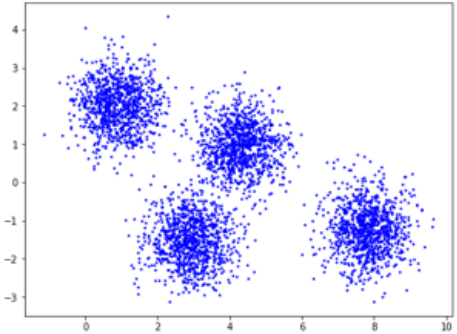
第 9 章 聚类算法

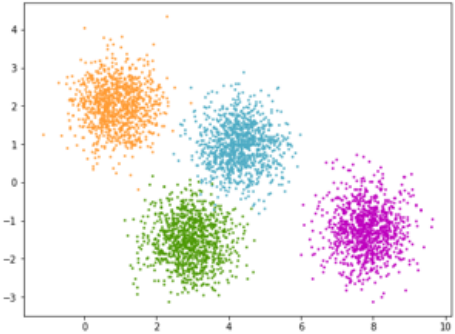
詹大为

计算机与人工智能学院

zhandawei@swjtu.edu.cn

1. 聚类任务

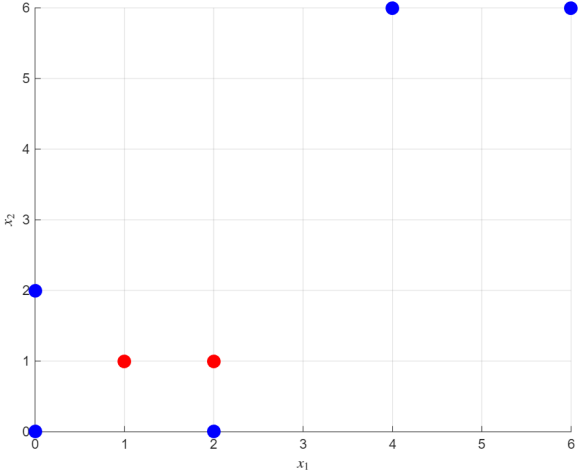




- 将 m 个无标签样本 $\mathcal{D}=\{\boldsymbol{x}_1,\boldsymbol{x}_2,\cdots,\boldsymbol{x}_m\}$ 划分为 k 个不相交的簇 $\mathcal{C}_i$，满足 $\cup_{i=1}^{k}\mathcal{C}_i=\mathcal{D}$，$\mathcal{C}_i\cap\mathcal{C}_j=\emptyset$，$i,j=1,2,\cdots,k$，$i\neq j$。
- 聚类结果通常用 $\{\lambda_1,\lambda_2,\cdots,\lambda_m\}$，$\lambda_i$ 为 $\boldsymbol{x}_i$ 的簇标签。

2.k 均值聚类算法：练习

现有 5 个样本 $\boldsymbol{x}_1 = [0,0]^{\mathrm{T}}, \boldsymbol{x}_2 = [0,2]^{\mathrm{T}}, \boldsymbol{x}_3 = [2,0]^{\mathrm{T}}, \boldsymbol{x}_4 = [4,6]^{\mathrm{T}}, \boldsymbol{x}_5 = [6,6]^{\mathrm{T}}$，使用 k 均值方法将 5 个样本聚成 2 个簇。初始聚类中心选为 $\boldsymbol{\mu}_1 = [1,1]^{\mathrm{T}}$ 和 $\boldsymbol{\mu}_2 = [2,1]^{\mathrm{T}}$。



2.k 均值聚类算法：练习

现有 5 个样本 $\boldsymbol{x}_1=[0,0]^{\mathrm{T}},\boldsymbol{x}_2=[0,2]^{\mathrm{T}},\boldsymbol{x}_3=[2,0]^{\mathrm{T}},\boldsymbol{x}_4=[4,6]^{\mathrm{T}},\boldsymbol{x}_5=[6,6]^{\mathrm{T}}$，使用 k 均值方法将 5 个样本聚成 2 个簇。初始聚类中心选为 $\boldsymbol{\mu}_1=[1,1]^{\mathrm{T}}$ 和 $\boldsymbol{\mu}_2=[2,1]^{\mathrm{T}}$。

初始聚类中心 $\boldsymbol{\mu}_1=[1,1]^{\mathrm{T}}$ 和 $\boldsymbol{\mu}_2=[2,1]^{\mathrm{T}}$。

$d(\boldsymbol{x}_1,\boldsymbol{\mu}_1)=\sqrt{2},\quad d(\boldsymbol{x}_1,\boldsymbol{\mu}_2)=\sqrt{5},\quad d(\boldsymbol{x}_1,\boldsymbol{\mu}_1)<d(\boldsymbol{x}_1,\boldsymbol{\mu}_2)$，因此 $\boldsymbol{x}_1\in\mathcal{C}_1$。

$d(\boldsymbol{x}_2,\boldsymbol{\mu}_1)=\sqrt{2},\quad d(\boldsymbol{x}_2,\boldsymbol{\mu}_2)=\sqrt{5},\quad d(\boldsymbol{x}_2,\boldsymbol{\mu}_1)<d(\boldsymbol{x}_2,\boldsymbol{\mu}_2)$，因此 $\boldsymbol{x}_2\in\mathcal{C}_1$。

$d(\boldsymbol{x}_3,\boldsymbol{\mu}_1)=\sqrt{2},\quad d(\boldsymbol{x}_3,\boldsymbol{\mu}_2)=\sqrt{1},\quad d(\boldsymbol{x}_3,\boldsymbol{\mu}_2)<d(\boldsymbol{x}_3,\boldsymbol{\mu}_1)$，因此 $\boldsymbol{x}_3\in\mathcal{C}_2$。

$d(\boldsymbol{x}_4,\boldsymbol{\mu}_1)=\sqrt{34},\quad d(\boldsymbol{x}_4,\boldsymbol{\mu}_2)=\sqrt{29},\quad d(\boldsymbol{x}_4,\boldsymbol{\mu}_2)<d(\boldsymbol{x}_4,\boldsymbol{\mu}_1)$，因此 $\boldsymbol{x}_4\in\mathcal{C}_2$。

$d(\boldsymbol{x}_5,\boldsymbol{\mu}_1)=\sqrt{50},\quad d(\boldsymbol{x}_5,\boldsymbol{\mu}_2)=\sqrt{41},\quad d(\boldsymbol{x}_5,\boldsymbol{\mu}_2)<d(\boldsymbol{x}_5,\boldsymbol{\mu}_1)$，因此 $\boldsymbol{x}_5\in\mathcal{C}_2$。

第 1 轮迭代聚类结果为 $\mathcal{C}_1=\{\boldsymbol{x}_1,\boldsymbol{x}_2\}$，$\mathcal{C}_2=\{\boldsymbol{x}_3,\boldsymbol{x}_4,\boldsymbol{x}_5\}$。

更新聚类中心 $\boldsymbol{\mu}_1=\frac{1}{2}\boldsymbol{x}_1+\frac{1}{2}\boldsymbol{x}_2=[0,1]^{\mathrm{T}}$，$\boldsymbol{\mu}_2=\frac{1}{3}\boldsymbol{x}_3+\frac{1}{3}\boldsymbol{x}_4+\frac{1}{3}\boldsymbol{x}_5=[4,4]^{\mathrm{T}}$。

2.k 均值聚类算法：练习

现有 5 个样本 $\boldsymbol{x}_1=[0,0]^{\mathrm{T}},\boldsymbol{x}_2=[0,2]^{\mathrm{T}},\boldsymbol{x}_3=[2,0]^{\mathrm{T}},\boldsymbol{x}_4=[4,6]^{\mathrm{T}},\boldsymbol{x}_5=[6,6]^{\mathrm{T}}$，使用 k 均值方法将 5 个样本聚成 2 个簇。初始聚类中心选为 $\boldsymbol{\mu}_1=[1,1]^{\mathrm{T}}$ 和 $\boldsymbol{\mu}_2=[2,1]^{\mathrm{T}}$。

第 2 轮迭代

$d(\boldsymbol{x}_1,\boldsymbol{\mu}_1)=\sqrt{1},\quad d(\boldsymbol{x}_1,\boldsymbol{\mu}_2)=\sqrt{32},\quad d(\boldsymbol{x}_1,\boldsymbol{\mu}_1)<d(\boldsymbol{x}_1,\boldsymbol{\mu}_2)$，因此 $\boldsymbol{x}_1\in\mathcal{C}_1$。

$d(\boldsymbol{x}_2,\boldsymbol{\mu}_1)=\sqrt{1},\quad d(\boldsymbol{x}_2,\boldsymbol{\mu}_2)=\sqrt{20},\quad d(\boldsymbol{x}_2,\boldsymbol{\mu}_1)<d(\boldsymbol{x}_2,\boldsymbol{\mu}_2)$，因此 $\boldsymbol{x}_2\in\mathcal{C}_1$。

$d(\boldsymbol{x}_3,\boldsymbol{\mu}_1)=\sqrt{2},\quad d(\boldsymbol{x}_3,\boldsymbol{\mu}_2)=\sqrt{20},\quad d(\boldsymbol{x}_3,\boldsymbol{\mu}_1)<d(\boldsymbol{x}_3,\boldsymbol{\mu}_2)$，因此 $\boldsymbol{x}_3\in\mathcal{C}_1$。

$d(\boldsymbol{x}_4,\boldsymbol{\mu}_1)=\sqrt{41},\quad d(\boldsymbol{x}_4,\boldsymbol{\mu}_2)=\sqrt{4},\quad d(\boldsymbol{x}_4,\boldsymbol{\mu}_2)<d(\boldsymbol{x}_4,\boldsymbol{\mu}_1)$，因此 $\boldsymbol{x}_4\in\mathcal{C}_2$。

$d(\boldsymbol{x}_5,\boldsymbol{\mu}_1)=\sqrt{61},\quad d(\boldsymbol{x}_5,\boldsymbol{\mu}_2)=\sqrt{8},\quad d(\boldsymbol{x}_5,\boldsymbol{\mu}_2)<d(\boldsymbol{x}_5,\boldsymbol{\mu}_1)$，因此 $\boldsymbol{x}_5\in\mathcal{C}_2$。

第 2 轮迭代聚类结果为 $\mathcal{C}_1=\{\boldsymbol{x}_1,\boldsymbol{x}_2,\boldsymbol{x}_3\}$，$\mathcal{C}_2=\{\boldsymbol{x}_4,\boldsymbol{x}_5\}$。

更新聚类中心 $\boldsymbol{\mu}_1=\frac{1}{3}\boldsymbol{x}_1+\frac{1}{3}\boldsymbol{x}_2+\frac{1}{3}\boldsymbol{x}_3=[0.67,0.67]^{\mathrm{T}}$，$\boldsymbol{\mu}_2=\frac{1}{2}\boldsymbol{x}_4+\frac{1}{2}\boldsymbol{x}_5=[5,6]^{\mathrm{T}}$。

2.k 均值聚类算法：练习

现有 5 个样本 $\boldsymbol{x}_1=[0,0]^{\mathrm{T}},\boldsymbol{x}_2=[0,2]^{\mathrm{T}},\boldsymbol{x}_3=[2,0]^{\mathrm{T}},\boldsymbol{x}_4=[4,6]^{\mathrm{T}},\boldsymbol{x}_5=[6,6]^{\mathrm{T}}$，使用 k 均值方法将 5 个样本聚成 2 个簇。初始聚类中心选为 $\boldsymbol{\mu}_1=[1,1]^{\mathrm{T}}$ 和 $\boldsymbol{\mu}_2=[2,1]^{\mathrm{T}}$。

第 3 轮迭代

$d(\boldsymbol{x}_1,\boldsymbol{\mu}_1)=\sqrt{0.898},\quad d(\boldsymbol{x}_1,\boldsymbol{\mu}_2)=\sqrt{61},\quad d(\boldsymbol{x}_1,\boldsymbol{\mu}_1)<d(\boldsymbol{x}_1,\boldsymbol{\mu}_2)$，因此 $\boldsymbol{x}_1\in\mathcal{C}_1$。

$d(\boldsymbol{x}_2,\boldsymbol{\mu}_1)=\sqrt{2.218},\quad d(\boldsymbol{x}_2,\boldsymbol{\mu}_2)=\sqrt{41},\quad d(\boldsymbol{x}_2,\boldsymbol{\mu}_1)<d(\boldsymbol{x}_2,\boldsymbol{\mu}_2)$，因此 $\boldsymbol{x}_2\in\mathcal{C}_1$。

$d(\boldsymbol{x}_3,\boldsymbol{\mu}_1)=\sqrt{2.218},\quad d(\boldsymbol{x}_3,\boldsymbol{\mu}_2)=\sqrt{45},\quad d(\boldsymbol{x}_3,\boldsymbol{\mu}_1)<d(\boldsymbol{x}_3,\boldsymbol{\mu}_2)$，因此 $\boldsymbol{x}_3\in\mathcal{C}_1$。

$d(\boldsymbol{x}_4,\boldsymbol{\mu}_1)=\sqrt{39.5},\quad d(\boldsymbol{x}_4,\boldsymbol{\mu}_2)=\sqrt{1},\quad d(\boldsymbol{x}_4,\boldsymbol{\mu}_2)<d(\boldsymbol{x}_4,\boldsymbol{\mu}_1)$，因此 $\boldsymbol{x}_4\in\mathcal{C}_2$。

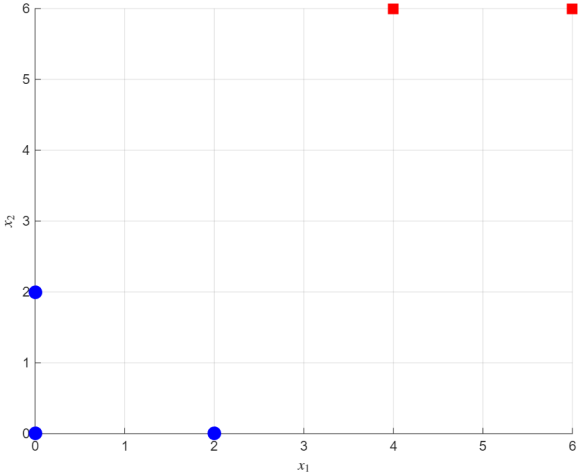
$d(\boldsymbol{x}_5,\boldsymbol{\mu}_1)=\sqrt{56.8},\quad d(\boldsymbol{x}_5,\boldsymbol{\mu}_2)=\sqrt{1},\quad d(\boldsymbol{x}_5,\boldsymbol{\mu}_2)<d(\boldsymbol{x}_5,\boldsymbol{\mu}_1)$，因此 $\boldsymbol{x}_5\in\mathcal{C}_2$。

第 3 轮迭代聚类结果为 $\mathcal{C}_1=\{\boldsymbol{x}_1,\boldsymbol{x}_2,\boldsymbol{x}_3\}$，$\mathcal{C}_2=\{\boldsymbol{x}_4,\boldsymbol{x}_5\}$。

聚类结果没有发生变化，聚类中心也不会变化，聚类过程收敛了。

2.k 均值聚类算法：练习

现有 5 个样本 $\boldsymbol{x}_1 = [0,0]^{\mathrm{T}}, \boldsymbol{x}_2 = [0,2]^{\mathrm{T}}, \boldsymbol{x}_3 = [2,0]^{\mathrm{T}}, \boldsymbol{x}_4 = [4,6]^{\mathrm{T}}, \boldsymbol{x}_5 = [6,6]^{\mathrm{T}}$，使用 k 均值方法将 5 个样本聚成 2 个簇。初始聚类中心选为 $\boldsymbol{\mu}_1 = [1,1]^{\mathrm{T}}$ 和 $\boldsymbol{\mu}_2 = [2,1]^{\mathrm{T}}$。



2.k 均值聚类算法

每个样本的簇标签依次为 $\boldsymbol{\lambda} = [\lambda_1, \lambda_2, \cdots, \lambda_m]$，$\lambda_i \in \{1, 2, \cdots, k\}$，优化问题：

$$\text{minimize: } E(\boldsymbol{\lambda}) = \sum_{i=1}^{k} \sum_{\boldsymbol{x} \in \mathcal{C}_i} ||\boldsymbol{x} - \boldsymbol{\mu}_i||_2^2, \quad \boldsymbol{\mu}_i = \frac{1}{|\mathcal{C}_i|} \sum_{\boldsymbol{x} \in \mathcal{C}_i} \boldsymbol{x}$$

- 损失函数求解是一个 NP 难问题。将 m 个样本分成 k 个簇，所有分法的数量为

$$N = \frac{1}{k!} \sum_{i=1}^{k} (-1)^{k-1} C_k^i k^m$$

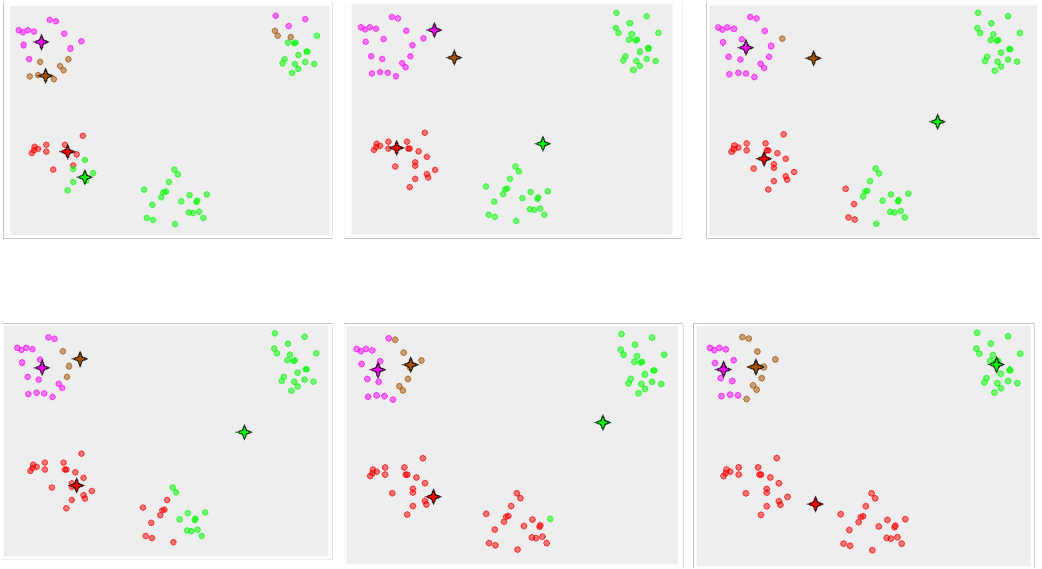
- k 均值采用贪心方法近似求解损失函数。

2.k 均值聚类算法

Algorithm 1 k 均值聚类算法

1: 随机选取 k 个点作为初始聚类中心 $\{\boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \cdots, \boldsymbol{\mu}_k\}$。
2: **repeat**
3: 　　**分配簇标签：** 以 $\{\boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \cdots, \boldsymbol{\mu}_k\}$ 为中心对所有样本聚类 $\{\mathcal{C}_i, \mathcal{C}_2, \cdots, \mathcal{C}_k\}$。
4: 　　**更新聚类中心：** 以聚类结果 $\{\mathcal{C}_i, \mathcal{C}_2, \cdots, \mathcal{C}_k\}$ 更新聚类中心 $\{\boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \cdots, \boldsymbol{\mu}_k\}$。
5: **until** 聚类中心 $\{\boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \cdots, \boldsymbol{\mu}_k\}$ 不发生变化
6: 输出聚类结果 $\{\mathcal{C}_1, \mathcal{C}_2, \cdots, \mathcal{C}_k\}$

2.k 均值聚类算法

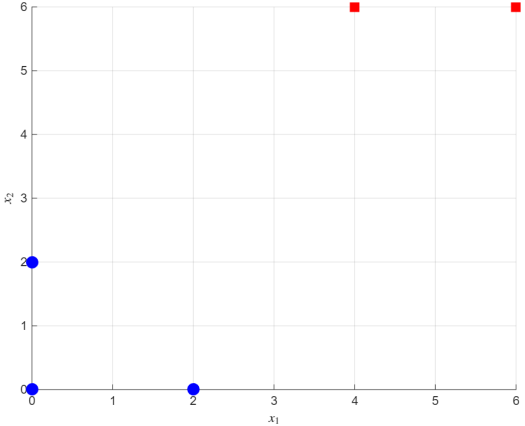


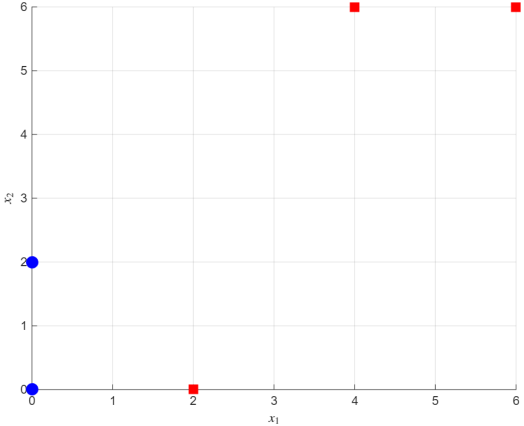
2.k 均值聚类算法：改进

1. **k-means++：**选取更好的聚类中心提高算法稳定性和收敛速度。
2. **k-medians：**选用曼哈顿距离度量，选取中值点进行聚类，减少外点对聚类结果的影响。

3. 聚类结果评价指标：外部指标

现有 5 个样本 $\boldsymbol{x}_1=[0,0]^{\mathrm{T}},\boldsymbol{x}_2=[0,2]^{\mathrm{T}},\boldsymbol{x}_3=[2,0]^{\mathrm{T}},\boldsymbol{x}_4=[4,6]^{\mathrm{T}},\boldsymbol{x}_5=[6,6]^{\mathrm{T}}$，使用 k 均值方法将 5 个样本聚成 2 个簇，结果为 $\mathcal{C}_1=\{\boldsymbol{x}_1,\boldsymbol{x}_2,\boldsymbol{x}_3\}$，$\mathcal{C}_2=\{\boldsymbol{x}_4,\boldsymbol{x}_5\}$。假设真实聚类结果为 $\mathcal{C}_1=\{\boldsymbol{x}_1,\boldsymbol{x}_2\}$，$\mathcal{C}_2=\{\boldsymbol{x}_3,\boldsymbol{x}_4,\boldsymbol{x}_5\}$，计算聚类结果的 Jaccard 系数、FM 指数、Rand 指数。





2. 聚类结果评价指标：外部指标

两个不同样本 $\boldsymbol{x}_i$ 和 $\boldsymbol{x}_j$，聚类簇标签为 λ_i 和 λ_j，真实簇标签为 $\lambda_i^\star$ 和 $\lambda_j^\star$

$$
\begin{aligned}
SS &= \{(\boldsymbol{x}_i, \boldsymbol{x}_j) | \lambda_i = \lambda_j, \lambda_i^\star = \lambda_j^\star, i < j\}, \quad a = |SS| \\
SD &= \{(\boldsymbol{x}_i, \boldsymbol{x}_j) | \lambda_i = \lambda_j, \lambda_i^\star \neq \lambda_j^\star, i < j\}, \quad b = |SD| \\
DS &= \{(\boldsymbol{x}_i, \boldsymbol{x}_j) | \lambda_i \neq \lambda_j, \lambda_i^\star = \lambda_j^\star, i < j\}, \quad c = |DS| \\
DD &= \{(\boldsymbol{x}_i, \boldsymbol{x}_j) | \lambda_i \neq \lambda_j, \lambda_i^\star \neq \lambda_j^\star, i < j\}, \quad d = |DD|
\end{aligned}
$$

总样本对数量

$$
a + b + c + d = \frac{m(m-1)}{2}
$$

2. 聚类结果评价指标：外部指标

Jaccard 系数（Jaccard Coefficient）

$$\mathrm{JC} = \frac{a}{a+b+c}$$

FM 指数（Fowlkes and Mallows Index）

$$\mathrm{FMI} = \frac{a}{\sqrt{(a+b)(a+c)}}$$

Rand 指数（Rand Index）

$$\mathrm{RI} = \frac{2(a+d)}{m(m-1)}$$

2. 聚类结果评价指标：练习

现有 5 个样本 $\boldsymbol{x}_1=[0,0]^{\mathrm{T}},\boldsymbol{x}_2=[0,2]^{\mathrm{T}},\boldsymbol{x}_3=[2,0]^{\mathrm{T}},\boldsymbol{x}_4=[4,6]^{\mathrm{T}},\boldsymbol{x}_5=[6,6]^{\mathrm{T}}$，使用 k 均值方法将 5 个样本聚成 2 个簇，结果为 $\mathcal{C}_1=\{\boldsymbol{x}_1,\boldsymbol{x}_2,\boldsymbol{x}_3\}$，$\mathcal{C}_2=\{\boldsymbol{x}_4,\boldsymbol{x}_5\}$。假设真实聚类结果为 $\mathcal{C}_1=\{\boldsymbol{x}_1,\boldsymbol{x}_2\}$，$\mathcal{C}_2=\{\boldsymbol{x}_3,\boldsymbol{x}_4,\boldsymbol{x}_5\}$，计算聚类结果的 Jaccard 系数、FM 指数、Rand 指数。

5 个样本所有样本对的全组合数为 $\frac{5\times4}{2}=10$，依次考虑每个组合属于哪一种情况

$\{\boldsymbol{x}_1,\boldsymbol{x}_2\}$ 属于 SS。

$\{\boldsymbol{x}_1,\boldsymbol{x}_3\}$ 属于 SD。

$\{\boldsymbol{x}_1,\boldsymbol{x}_4\}$ 属于 DD。

$\{\boldsymbol{x}_1,\boldsymbol{x}_5\}$ 属于 DD。

$\{\boldsymbol{x}_2,\boldsymbol{x}_3\}$ 属于 SD。

$\{\boldsymbol{x}_2,\boldsymbol{x}_4\}$ 属于 DD。

$\{\boldsymbol{x}_2,\boldsymbol{x}_5\}$ 属于 DD。

$\{\boldsymbol{x}_3,\boldsymbol{x}_4\}$ 属于 DS。

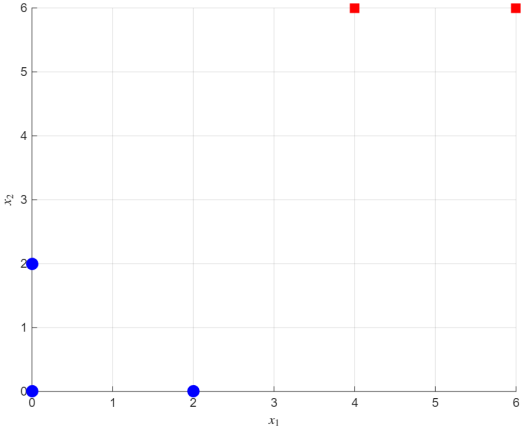
$\{\boldsymbol{x}_3,\boldsymbol{x}_5\}$ 属于 DS。

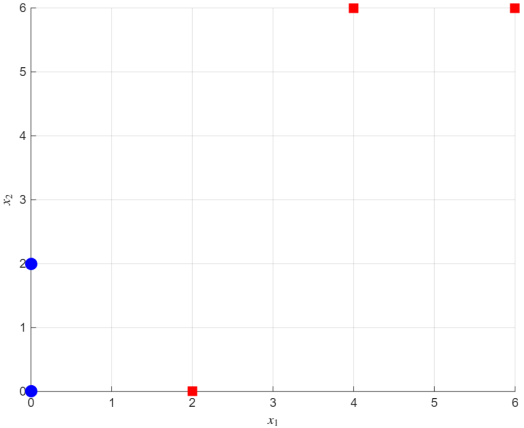
$\{\boldsymbol{x}_4,\boldsymbol{x}_5\}$ 属于 SS。

$$a=|SS|=2$$
$$b=|SD|=2$$
$$c=|DS|=2$$
$$d=|DD|=4$$
$$\mathrm{JC}=\frac{a}{a+b+c}=\frac{2}{6}=0.33$$
$$\mathrm{FMI}=\frac{a}{\sqrt{(a+b)(a+c)}}=\frac{2}{\sqrt{4\times4}}=0.5$$
$$\mathrm{RI}=\frac{2(a+d)}{m(m-1)}=\frac{12}{20}=0.6$$

3. 聚类结果评价指标：内部指标

现有 5 个样本 $\boldsymbol{x}_1=[0,0]^{\mathrm{T}},\boldsymbol{x}_2=[0,2]^{\mathrm{T}},\boldsymbol{x}_3=[2,0]^{\mathrm{T}},\boldsymbol{x}_4=[4,6]^{\mathrm{T}},\boldsymbol{x}_5=[6,6]^{\mathrm{T}}$。聚类算法 1 的聚类结果为 $\mathcal{C}_1=\{\boldsymbol{x}_1,\boldsymbol{x}_2,\boldsymbol{x}_3\}$，$\mathcal{C}_2=\{\boldsymbol{x}_4,\boldsymbol{x}_5\}$。聚类算法 2 的聚类结果为 $\mathcal{C}_1=\{\boldsymbol{x}_1,\boldsymbol{x}_2\}$，$\mathcal{C}_2=\{\boldsymbol{x}_3,\boldsymbol{x}_4,\boldsymbol{x}_5\}$，计算两个聚类结果的 Dunn 指数。





2. 聚类结果评价指标：内部指标

簇内平均距离

$$\mathrm{avg}(\mathcal{C}) = \frac{2}{|\mathcal{C}|(|\mathcal{C}|-1)} \sum_{1 \le i < j \le |\mathcal{C}|} d(\boldsymbol{x}_i, \boldsymbol{x}_j)$$

簇内最大距离（簇直径）

$$\mathrm{diam}(\mathcal{C}) = \max_{1 \le i < j \le |\mathcal{C}|} d(\boldsymbol{x}_i, \boldsymbol{x}_j)$$

簇间最小距离

$$d_{\min}(\mathcal{C}_i, \mathcal{C}_j) = \min_{\boldsymbol{x}_p \in \mathcal{C}_i, \boldsymbol{x}_q \in \mathcal{C}_j} d(\boldsymbol{x}_p, \boldsymbol{x}_q)$$

簇间中心距离

$$d_{\mathrm{cen}}(\mathcal{C}_i, \mathcal{C}_j) = d(\boldsymbol{\mu}_i, \boldsymbol{\mu}_j), \quad \boldsymbol{\mu}_i = \frac{1}{|\mathcal{C}_i|} \sum_{\boldsymbol{x} \in \mathcal{C}_i} \boldsymbol{x}, \boldsymbol{\mu}_j = \frac{1}{|\mathcal{C}_j|} \sum_{\boldsymbol{x} \in \mathcal{C}_j} \boldsymbol{x}$$

2. 聚类结果评价指标：内部指标

DB 指数（Davies Bouldin Index）

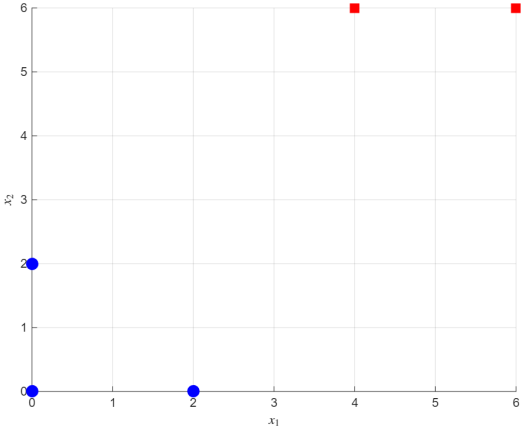
$$\mathrm{DBI} = \frac{1}{k}\sum_{i=1}^{k} \max_{j=1,2,\cdots,k,j\neq i} \left(\frac{\mathrm{avg}(\mathcal{C}_i) + \mathrm{avg}(\mathcal{C}_j)}{d_{\mathrm{cen}}(\mathcal{C}_i, \mathcal{C}_j)} \right)$$

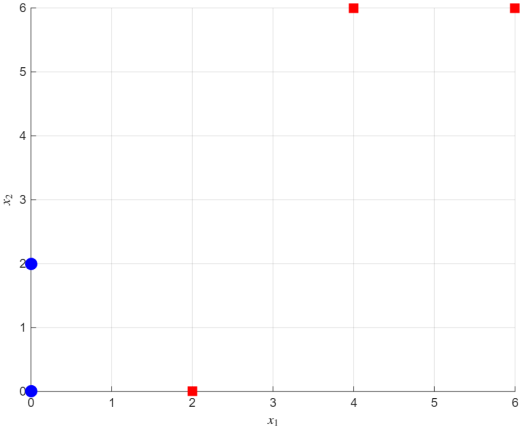
Dunn 指数（Dunn Index）

$$\mathrm{DI} = \frac{\min\limits_{1\leq i<j\leq k} d_{\min}(\mathcal{C}_i, \mathcal{C}_j)}{\max\limits_{1\leq i\leq k} \mathrm{diam}(\mathcal{C}_i)}$$

3. 聚类结果评价指标：内部指标

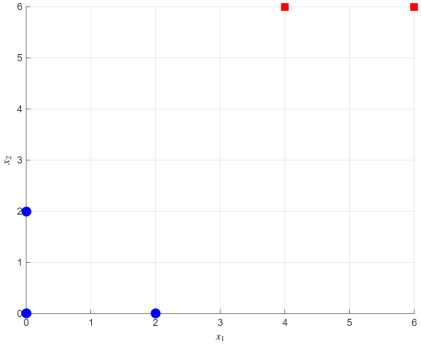
现有 5 个样本 $\boldsymbol{x}_1=[0,0]^{\mathrm{T}},\boldsymbol{x}_2=[0,2]^{\mathrm{T}},\boldsymbol{x}_3=[2,0]^{\mathrm{T}},\boldsymbol{x}_4=[4,6]^{\mathrm{T}},\boldsymbol{x}_5=[6,6]^{\mathrm{T}}$。聚类算法 1 的聚类结果为 $\mathcal{C}_1=\{\boldsymbol{x}_1,\boldsymbol{x}_2,\boldsymbol{x}_3\}$，$\mathcal{C}_2=\{\boldsymbol{x}_4,\boldsymbol{x}_5\}$。聚类算法 2 的聚类结果为 $\mathcal{C}_1=\{\boldsymbol{x}_1,\boldsymbol{x}_2\}$，$\mathcal{C}_2=\{\boldsymbol{x}_3,\boldsymbol{x}_4,\boldsymbol{x}_5\}$，计算两个聚类结果的 Dunn 指数。





3. 聚类结果评价指标：内部指标

现有 5 个样本 $\boldsymbol{x}_1=[0,0]^{\mathrm{T}}, \boldsymbol{x}_2=[0,2]^{\mathrm{T}}, \boldsymbol{x}_3=[2,0]^{\mathrm{T}}, \boldsymbol{x}_4=[4,6]^{\mathrm{T}}, \boldsymbol{x}_5=[6,6]^{\mathrm{T}}$。聚类算法 1 的聚类结果为 $\mathcal{C}_1=\{\boldsymbol{x}_1,\boldsymbol{x}_2,\boldsymbol{x}_3\}$，$\mathcal{C}_2=\{\boldsymbol{x}_4,\boldsymbol{x}_5\}$。聚类算法 2 的聚类结果为 $\mathcal{C}_1=\{\boldsymbol{x}_1,\boldsymbol{x}_2\}$，$\mathcal{C}_2=\{\boldsymbol{x}_3,\boldsymbol{x}_4,\boldsymbol{x}_5\}$，计算两个聚类结果的 Dunn 指数。



聚类算法 1:

$$d_{\min}(\mathcal{C}_1,\mathcal{C}_2)=d(\boldsymbol{x}_2,\boldsymbol{x}_4)=\sqrt{32}$$

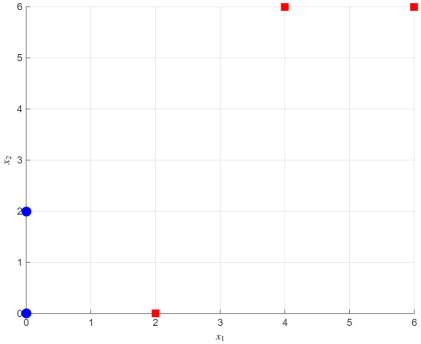
$$\mathrm{diam}(\mathcal{C}_1)=d(\boldsymbol{x}_2,\boldsymbol{x}_3)=\sqrt{8}$$

$$\mathrm{diam}(\mathcal{C}_2)=d(\boldsymbol{x}_4,\boldsymbol{x}_5)=2$$

$$\mathrm{DI}=\frac{d_{\min}(\mathcal{C}_1,\mathcal{C}_2)}{\max\limits_{1\le i\le 2}\mathrm{diam}(\mathcal{C}_i)}=\frac{d_{\min}(\mathcal{C}_1,\mathcal{C}_2)}{\mathrm{diam}(\mathcal{C}_1)}=\frac{\sqrt{32}}{\sqrt{8}}=2$$

3. 聚类结果评价指标：内部指标

现有 5 个样本 $\boldsymbol{x}_1=[0,0]^{\mathrm{T}}, \boldsymbol{x}_2=[0,2]^{\mathrm{T}}, \boldsymbol{x}_3=[2,0]^{\mathrm{T}}, \boldsymbol{x}_4=[4,6]^{\mathrm{T}}, \boldsymbol{x}_5=[6,6]^{\mathrm{T}}$。聚类算法 1 的聚类结果为 $C_1=\{\boldsymbol{x}_1,\boldsymbol{x}_2,\boldsymbol{x}_3\}$，$C_2=\{\boldsymbol{x}_4,\boldsymbol{x}_5\}$。聚类算法 2 的聚类结果为 $C_1=\{\boldsymbol{x}_1,\boldsymbol{x}_2\}$，$C_2=\{\boldsymbol{x}_3,\boldsymbol{x}_4,\boldsymbol{x}_5\}$，计算两个聚类结果的 Dunn 指数。



聚类算法 2：

$$d_{\min}(\mathcal{C}_1,\mathcal{C}_2)=d(\boldsymbol{x}_1,\boldsymbol{x}_3)=2$$

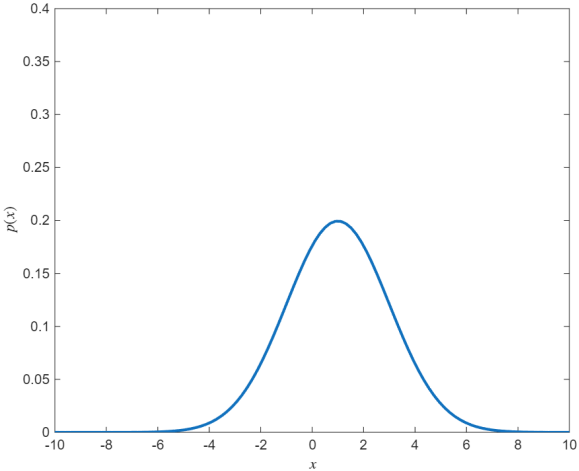
$$\mathrm{diam}(\mathcal{C}_1)=d(\boldsymbol{x}_1,\boldsymbol{x}_2)=2$$

$$\mathrm{diam}(\mathcal{C}_2)=d(\boldsymbol{x}_3,\boldsymbol{x}_5)=\sqrt{52}$$

$$\mathrm{DI}=\frac{d_{\min}(\mathcal{C}_1,\mathcal{C}_2)}{\max\limits_{1\le i\le 2}\mathrm{diam}(\mathcal{C}_i)}=\frac{d_{\min}(\mathcal{C}_1,\mathcal{C}_2)}{\mathrm{diam}(\mathcal{C}_2)}=\frac{2}{\sqrt{52}}\approx 0.28$$

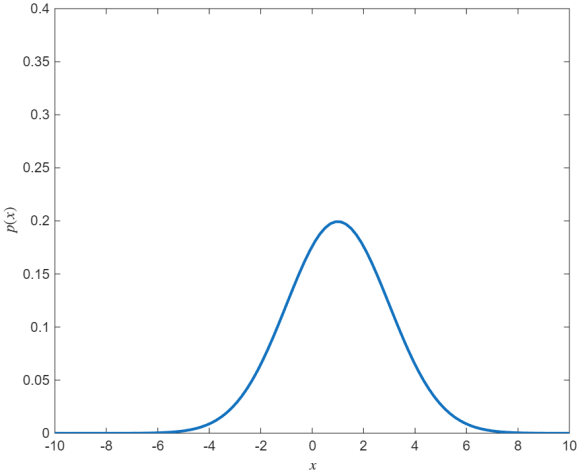
4. 高斯混合聚类：Gaussian Mixture Model

$$p(x|\mu,\sigma) = \frac{1}{\sqrt{2\pi}\sigma}\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$



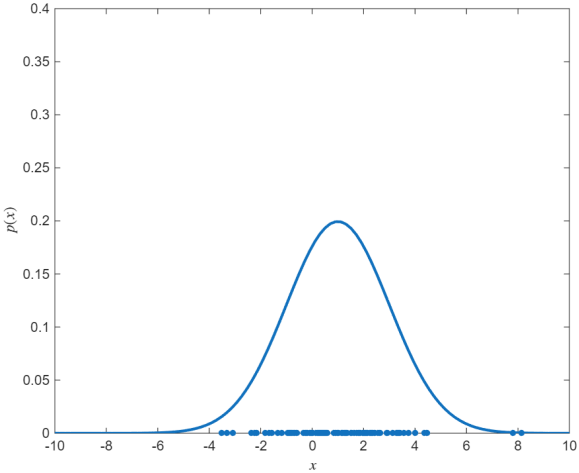
4. 高斯混合聚类：Gaussian Mixture Model

$$p(x|\mu_1 = 1, \sigma_1 = 2)$$



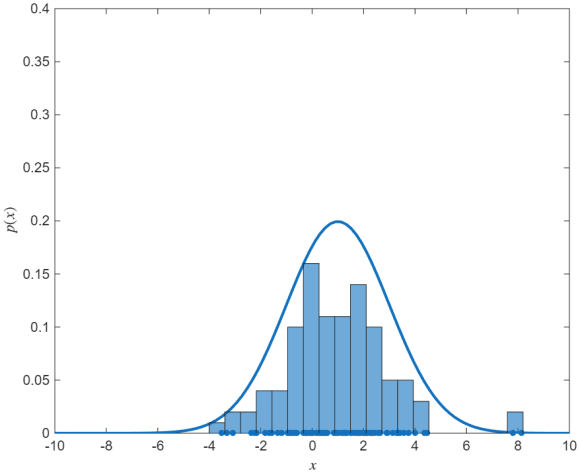
4. 高斯混合聚类：Gaussian Mixture Model

$$p(x|\mu_1 = 1, \sigma_1 = 2)$$



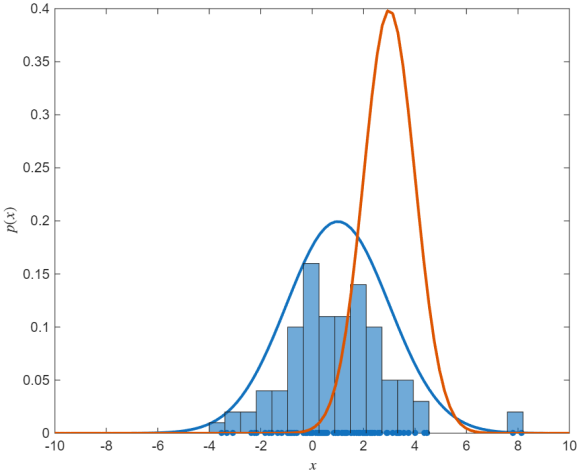
4. 高斯混合聚类：Gaussian Mixture Model

$$p(x|\mu_1 = 1, \sigma_1 = 2)$$



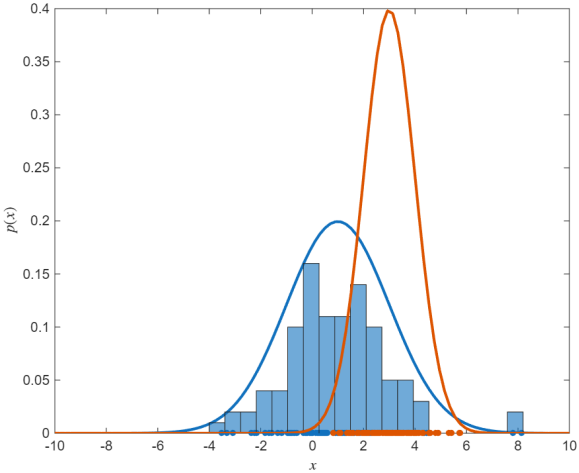
4. 高斯混合聚类：Gaussian Mixture Model

$$p(x|\mu_1 = 1, \sigma_1 = 2), \quad p(x|\mu_2 = 3, \sigma_2 = 1)$$



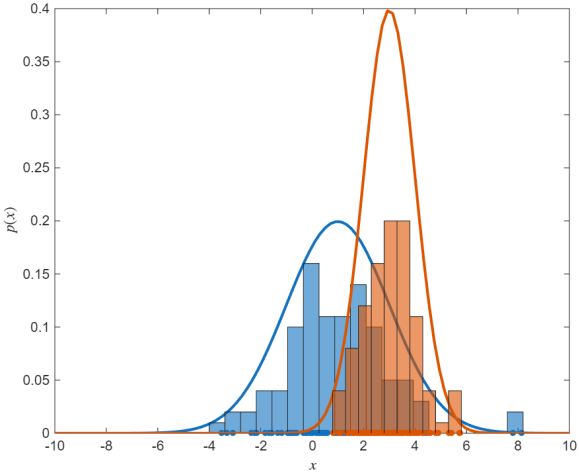
4. 高斯混合聚类：Gaussian Mixture Model

$$p(x|\mu_1 = 1, \sigma_1 = 2), \quad p(x|\mu_2 = 3, \sigma_2 = 1)$$



4. 高斯混合聚类：Gaussian Mixture Model

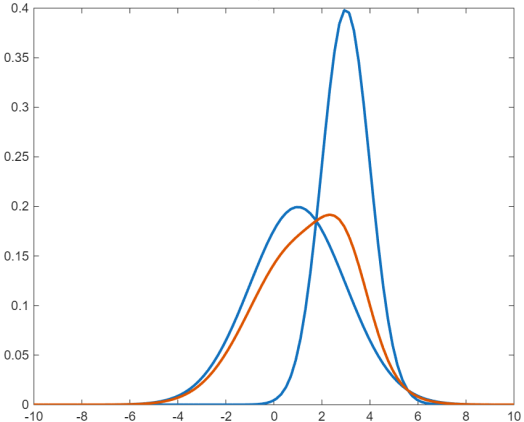
$$p(x|\mu_1 = 1, \sigma_1 = 2), \quad p(x|\mu_2 = 3, \sigma_2 = 1)$$



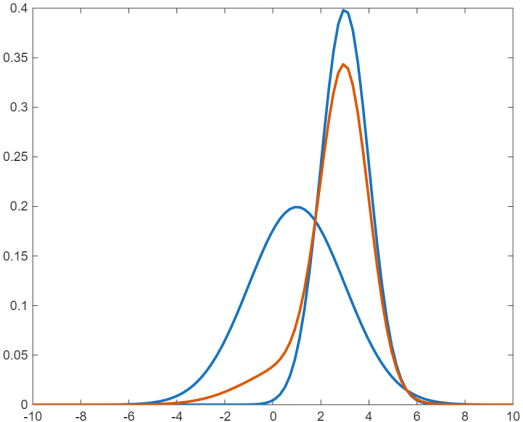
4. 高斯混合聚类：Gaussian Mixture Model

$$p_{\mathcal{M}}(x) = \alpha_1 p(x|\mu_1 = 1, \sigma_1 = 2) + \alpha_2 p(x|\mu_2 = 3, \sigma_2 = 1)$$

$\alpha_1 = 0.8, \alpha_2 = 0.2$



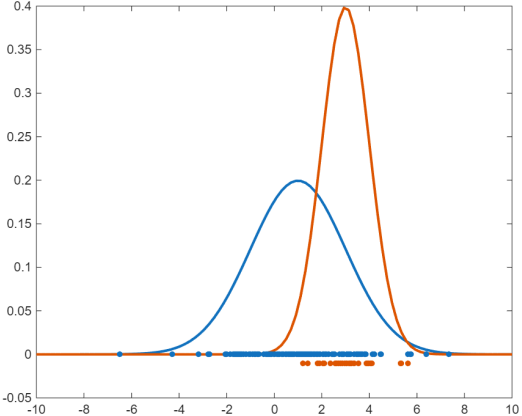
$\alpha_1 = 0.2, \alpha_2 = 0.8$



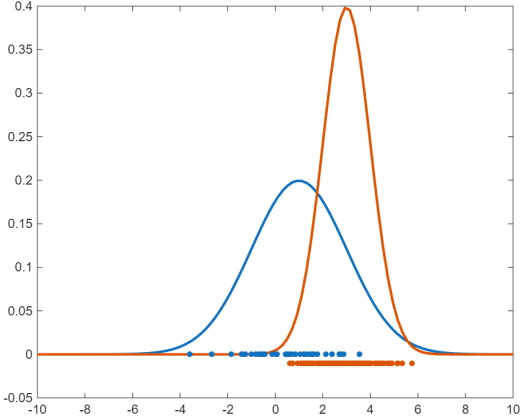
4. 高斯混合聚类：Gaussian Mixture Model

$$p_{\mathcal{M}}(x) = \alpha_1 p(x|\mu_1 = 1, \sigma_1 = 2) + \alpha_2 p(x|\mu_2 = 3, \sigma_2 = 1)$$

$\alpha_1 = 0.8, \alpha_2 = 0.2$



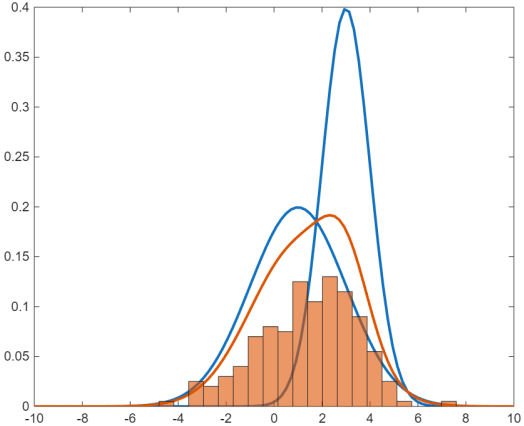
$\alpha_1 = 0.2, \alpha_2 = 0.8$



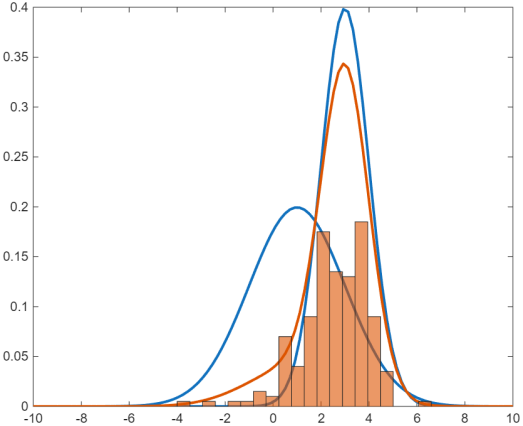
4. 高斯混合聚类：Gaussian Mixture Model

$$p_{\mathcal{M}}(x) = \alpha_1 p(x|\mu_1 = 1, \sigma_1 = 2) + \alpha_2 p(x|\mu_2 = 3, \sigma_2 = 1)$$

$\alpha_1 = 0.8, \alpha_2 = 0.2$

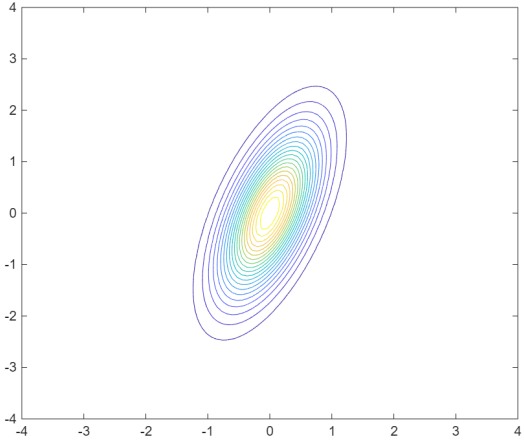


$\alpha_1 = 0.2, \alpha_2 = 0.8$



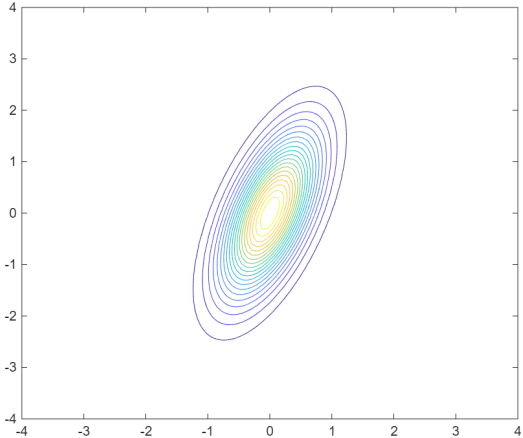
4. 高斯混合聚类：二元高斯分布

$$p(\boldsymbol{x}|\boldsymbol{\mu},\boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{\frac{d}{2}}}|\boldsymbol{\Sigma}|^{\frac{1}{2}}\exp\left(-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^{\mathrm{T}}\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})\right)$$



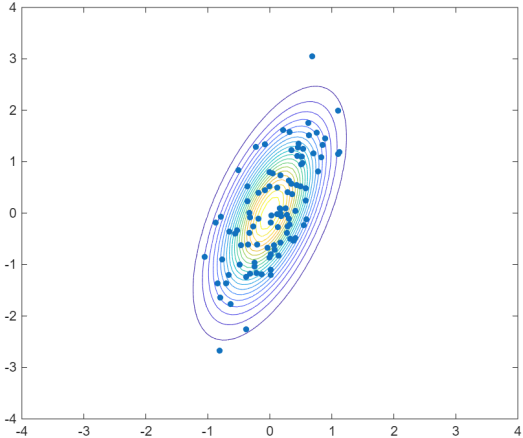
4. 高斯混合聚类：二元高斯分布

$$p\left(\boldsymbol{x}|\boldsymbol{\mu}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \boldsymbol{\Sigma}_1 = \begin{bmatrix} 0.25 & 0.3 \\ 0.3 & 1 \end{bmatrix}\right)$$



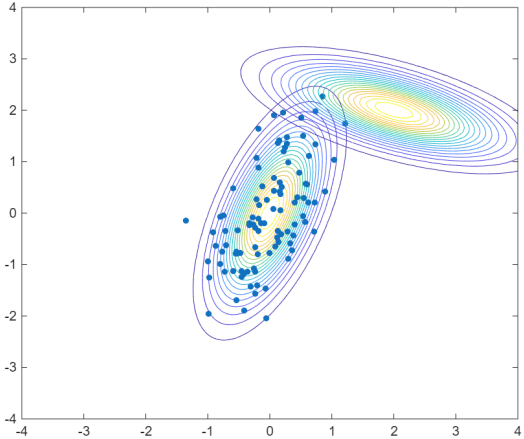
4. 高斯混合聚类：二元高斯分布

$$p\left(\boldsymbol{x}|\boldsymbol{\mu}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \boldsymbol{\Sigma}_1 = \begin{bmatrix} 0.25 & 0.3 \\ 0.3 & 1 \end{bmatrix}\right)$$



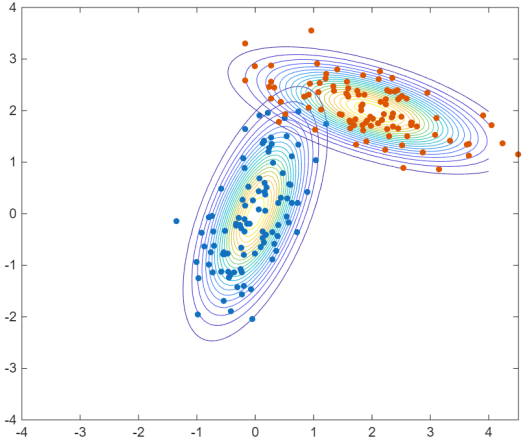
4. 高斯混合聚类：二元高斯分布

$$p\left(\boldsymbol{x}|\boldsymbol{\mu}_1=\begin{bmatrix}0\\0\end{bmatrix},\boldsymbol{\Sigma}_1=\begin{bmatrix}0.25 & 0.3\\0.3 & 1\end{bmatrix}\right),\quad p\left(\boldsymbol{x}|\boldsymbol{\mu}_2=\begin{bmatrix}2\\2\end{bmatrix},\boldsymbol{\Sigma}_2=\begin{bmatrix}1 & -0.3\\-0.3 & 0.25\end{bmatrix}\right)$$



4. 高斯混合聚类：二元高斯分布

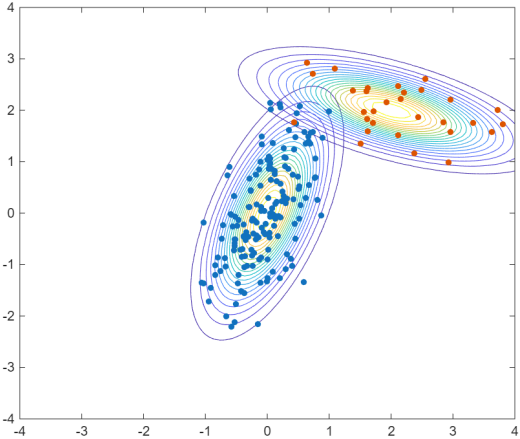
$$p\left(\boldsymbol{x}|\boldsymbol{\mu}_1=\begin{bmatrix}0\\0\end{bmatrix},\boldsymbol{\Sigma}_1=\begin{bmatrix}0.25 & 0.3\\0.3 & 1\end{bmatrix}\right),\quad p\left(\boldsymbol{x}|\boldsymbol{\mu}_2=\begin{bmatrix}2\\2\end{bmatrix},\boldsymbol{\Sigma}_2=\begin{bmatrix}1 & -0.3\\-0.3 & 0.25\end{bmatrix}\right)$$



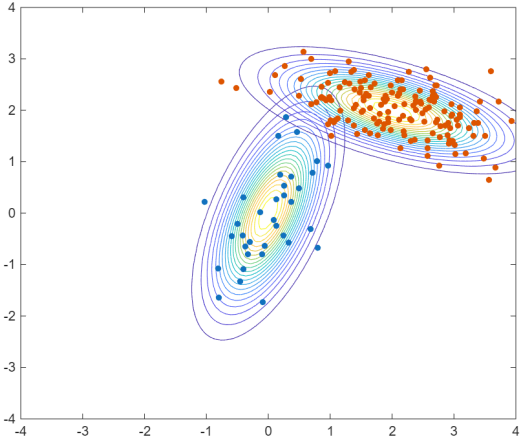
4. 高斯混合聚类：二元高斯分布

$$p_{\mathcal{M}}(\boldsymbol{x}) = \alpha_1 p(\boldsymbol{x}|\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1) + \alpha_2 p(\boldsymbol{x}|\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_2)$$

$\alpha_1 = 0.8, \alpha_2 = 0.2$

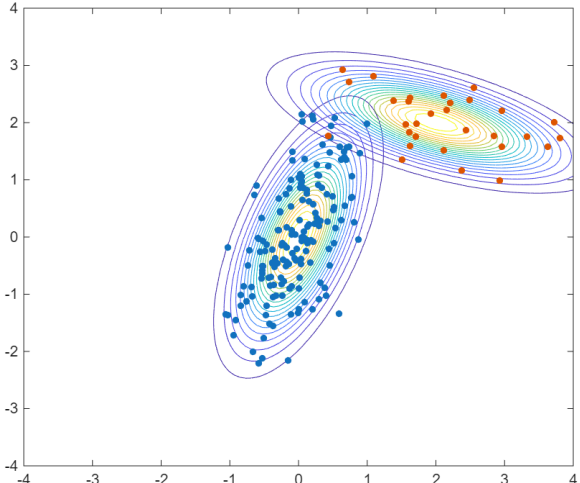


$\alpha_1 = 0.2, \alpha_2 = 0.8$

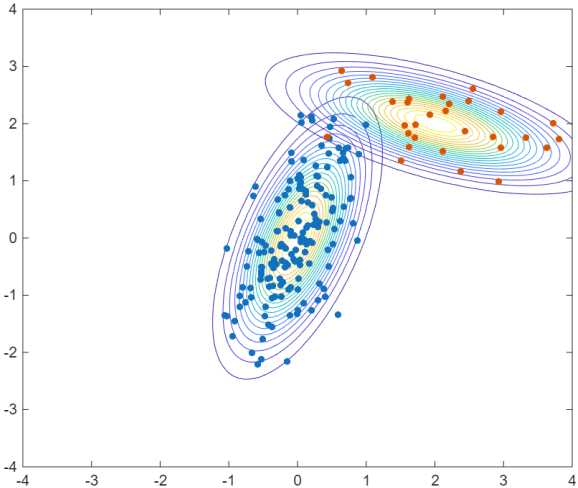


4. 高斯混合聚类：Gaussian Mixture Model

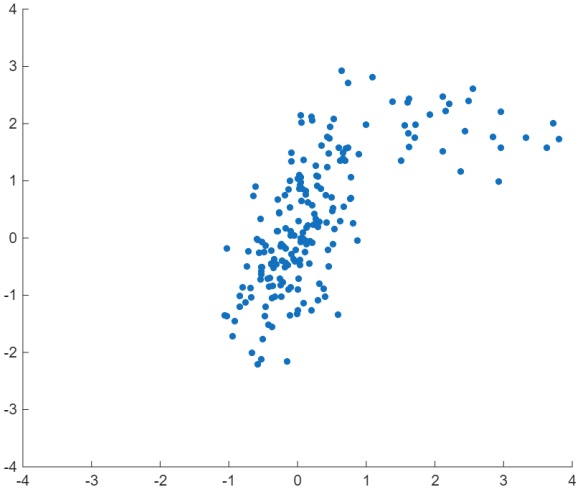
$$p\left(\boldsymbol{x}|\boldsymbol{\mu}_1=\begin{bmatrix}0\\0\end{bmatrix},\boldsymbol{\Sigma}_1=\begin{bmatrix}0.25 & 0.3\\0.3 & 1\end{bmatrix}\right),p\left(\boldsymbol{x}|\boldsymbol{\mu}_2=\begin{bmatrix}2\\2\end{bmatrix},\boldsymbol{\Sigma}_2=\begin{bmatrix}1 & -0.3\\-0.3 & 0.25\end{bmatrix}\right),\alpha_1=0.8,\alpha_2=0.2$$



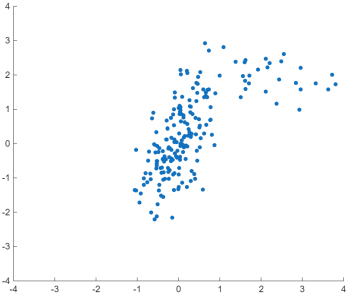
4. 高斯混合聚类：Gaussian Mixture Model



4. 高斯混合聚类：Gaussian Mixture Model



4. 高斯混合聚类：假设



假设数据由高斯混合分布采样得到，通过数据估计 α_i、$\boldsymbol{\mu}_i$、$\boldsymbol{\Sigma}_i$，则可以计算任意样本 $\boldsymbol{x}$ 是由第 i 个高斯分布生成的后验概率：

$$P(Z=i|\boldsymbol{x})=\frac{P(Z=i)p_{\mathcal{M}}(\boldsymbol{x}|Z=i)}{p_{\mathcal{M}}(\boldsymbol{x})}$$

计算出后验概率 $P(Z=1|\boldsymbol{x}),\cdots,P(Z=k|\boldsymbol{x})$ 后，预测 $\boldsymbol{x}$ 的簇标签：

$$\lambda=\max_{1\leq i\leq k}P(Z=i|\boldsymbol{x})$$

4. 高斯混合聚类：理论求解方法

混合模型概率密度函数

$$p_{\mathcal{M}}(\boldsymbol{x}) = \sum_{i=1}^{k} \alpha_i p(\boldsymbol{x}|\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)$$

将获取样本 $\{\boldsymbol{x}_1, \boldsymbol{x}_2, \cdots, \boldsymbol{x}_m\}$ 带入得到似然函数

$$L = \prod_{j=1}^{m} p_{\mathcal{M}}(\boldsymbol{x}_j) = \prod_{j=1}^{m} \sum_{i=1}^{k} \alpha_i p(\boldsymbol{x}_j|\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)$$

$$L(\alpha_1, \cdots, \alpha_k, \boldsymbol{\mu}_1, \cdots, \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_1, \cdots, \boldsymbol{\Sigma}_k) = \prod_{j=1}^{m} \sum_{i=1}^{k} \alpha_i p(\boldsymbol{x}_j|\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)$$

似然函数变量数目过多，难以求解，高斯混合聚类采用贪心方法近似求解。

4. 高斯混合聚类：算法流程

Algorithm 2 高斯混合聚类算法

初始化模型参数 α_i、$\boldsymbol{\mu}_i$、$\boldsymbol{\Sigma}_i$，$i = 1, 2, \cdots, k$

repeat

  1. 聚类：

  for $j = 1$ to m **do**

$$P(Z = i|\boldsymbol{x}_j) = \frac{P(Z = i)p_{\mathcal{M}}(\boldsymbol{x}_j|Z = i)}{p_{\mathcal{M}}(\boldsymbol{x}_j)}, i = 1, 2, \cdots, k$$

$$\lambda_j = \max_{1 \le i \le k} P(Z = i|\boldsymbol{x}_j)$$

  end for

  2. 更新参数： $\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i, \alpha_i = \frac{1}{m}\sum_{j=1}^{m} \lambda_j$，$i = 1, 2, \cdots, k$。

until 聚类结果不再发生变化

输出聚类结果 $\{\lambda_1, \lambda_2, \cdots, \lambda_m\}$。

4. 高斯混合聚类：各个概率计算方法

已知参数 α_i、$\boldsymbol{\mu}_i$、$\boldsymbol{\Sigma}_i$，$i = 1, 2, \cdots, k$

$$P(Z = i|\boldsymbol{x}) = \frac{P(Z = i)p_{\mathcal{M}}(\boldsymbol{x}|Z = i)}{p_{\mathcal{M}}(\boldsymbol{x})}$$

$$P(Z = i) = \alpha_i$$

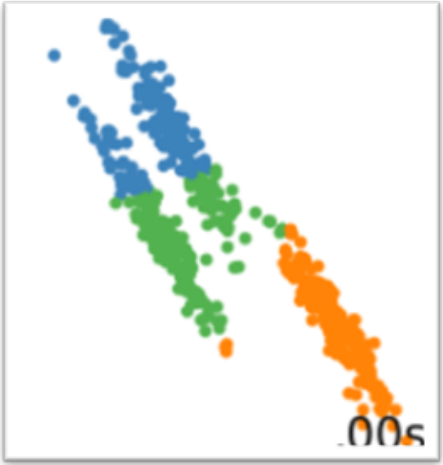
$$p_{\mathcal{M}}(\boldsymbol{x}|Z = i) = p(\boldsymbol{x}|\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)$$

$$p_{\mathcal{M}}(\boldsymbol{x}) = \sum_{i=1}^{k} \alpha_i p(\boldsymbol{x}|\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)$$

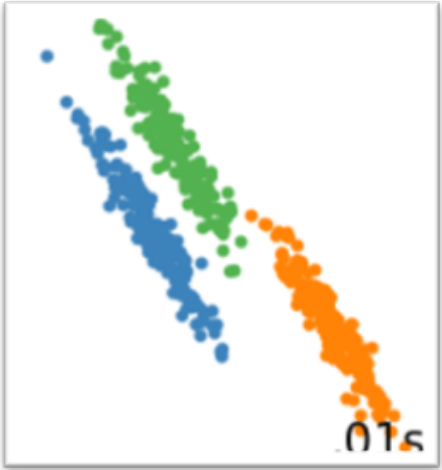
$$P(Z = i|\boldsymbol{x}) = \frac{\alpha_i p(\boldsymbol{x}|\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)}{\sum_{l=1}^{k} \alpha_l p(\boldsymbol{x}|\boldsymbol{\mu}_l, \boldsymbol{\Sigma}_l)}$$

4. 高斯混合聚类

k 均值聚类

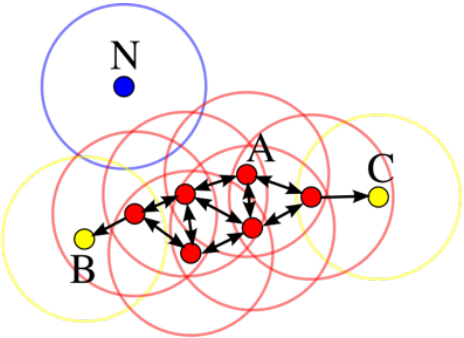


高斯混合聚类



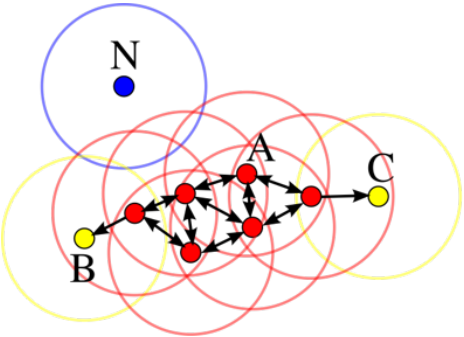
5. DNSCAN：密度聚类方法

DBSCAN: Density-Based Spatial Clustering of Applications with Noise



1. 核心点：邻域内至少包含 $minPts$ 个点。
2. 密度直达：$\boldsymbol{p}$ 为核心点，$\boldsymbol{q}$ 在 $\boldsymbol{p}$ 的邻域内，则称 $\boldsymbol{q}$ 由 $\boldsymbol{p}$ 密度直达。
3. 密度可达：$\boldsymbol{p}$ 为核心点，存在序列 $\boldsymbol{p}_1, \boldsymbol{p}_2, \cdots, \boldsymbol{p}_k$ 密度直达，称 $\boldsymbol{p}_k$ 由 $\boldsymbol{p}$ 密度可达。

5. DNSCAN：密度聚类方法



Algorithm 3 DBSCAN

遍历所有样本，找出所有核心点 $\mathcal{O} = \{\boldsymbol{o}_1, \boldsymbol{o}_2, \cdots, \boldsymbol{o}_t\}$

repeat

　　从 $\mathcal{O}$ 中随机选取一个核心点，找出它所有密度可达点，构成一个簇。

　　将已构成簇的所有核心点从集合中删除。

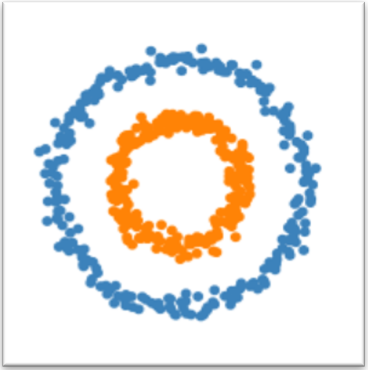
until $\mathcal{O} = \emptyset$

5. DNSCAN：密度聚类方法

k 均值聚类

高斯混合聚类

DBSCAN

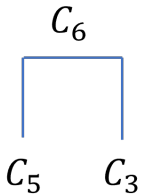


6. AGNES：Agglomerative Nesting 层次聚类方法

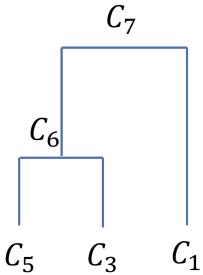
1. 每个样本初始化为一个单独的簇。
2. 找出距离最近的两个簇进行合并。
3. 直到簇的数量达到要求。

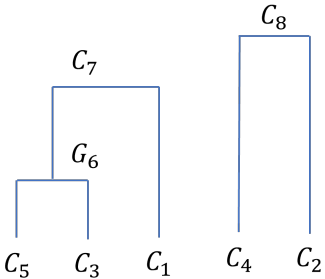
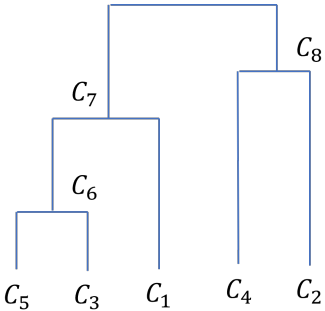
6. AGNES：层次聚类方法

$$
\begin{array}{c|ccccc}
 & C_1 & C_2 & C_3 & C_4 & C_5 \\
\hline
C_1 & 0 & 7 & 2 & 9 & 3 \\
C_2 & 7 & 0 & 5 & 4 & 6 \\
C_3 & 2 & 5 & 0 & 8 & 1 \\
C_4 & 9 & 4 & 8 & 0 & 5 \\
C_5 & 3 & 6 & 1 & 5 & 0
\end{array}
$$



$$
\begin{array}{c|cccc}
 & C_1 & C_2 & C_4 & C_6 \\
\hline
C_1 & 0 & 7 & 9 & 2 \\
C_2 & 7 & 0 & 4 & 5 \\
C_4 & 9 & 4 & 0 & 5 \\
C_6 & 2 & 5 & 5 & 0
\end{array}
$$



6. AGNES：层次聚类方法

$$
\begin{array}{c c}
 & \begin{array}{ccc} C_2 & C_4 & C_7 \end{array} \\
\begin{array}{c} C_2 \\ C_4 \\ C_7 \end{array} & \begin{bmatrix} 0 & 4 & 5 \\ 4 & 0 & 5 \\ 5 & 5 & 0 \end{bmatrix}
\end{array}
$$